ASSOCIATED PRIMES OF AN S-MODULE CONTRACT ALONG A RING MAP TO ASSOCIATED PRIMES OVER THE SOURCE RING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Characterisation of associated primes by annihilators of elements. Used at the start of the counted proof (lines 15553-15554). Turns membership of a prime in the associated primes into the existence of an element whose annihilator is that prime.
- Contraction of an ideal along a ring map. Used at the end of the counted proof (line 15555). Identifies the annihilator over R of the same element with the contraction of its annihilator over S.

Lemma 0.1. *Let $\varphi : R \to S$ be a ring map. Let M be an S-module. Then* $\mathrm{Spec}(\varphi)(Ass_S(M)) \subset Ass_R(M)$.

Proof. If $\mathfrak{q} \in \mathrm{Ass}_S(M)$, then there exists an m in M such that the annihilator of m in S is $\mathfrak{q}$. Then the annihilator of m in R is $\mathfrak{q} \cap R$. □

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